

ANSWERS

1. (e) 2. (e) 3. (e) 4. (d) 5. (e) 6. (e) 7. (b) 8. (c) 9. (d) 10. (e)
 11. (e) 12. (d) 13. (a) 14. (e) 15. (a) 16. (c)

SOLUTIONS

1. Given : $P + Q + R = 96$... (i)
 I. $P = R + 6$... (ii)
 II. $Q + R = 56$... (iii)
 On subtracting (iii) from (i), we get $P = 40$.
 Putting $P = 40$ in (ii), we get $R = 34$. Putting $R = 34$ in (iii), we get $Q = 22$.
 Thus, I and II both together give the answer. So, correct answer is (e).
2. I. $S = 5D \Rightarrow D = \frac{S}{5}$... (i)
 II. $S - 5 = 25(D - 5) \Leftrightarrow S = 25D - 120$... (ii)
 Using (i) in (ii), we get
 $S = \left(25 \times \frac{S}{5}\right) - 120 \Leftrightarrow 4S = 120 \Leftrightarrow S = 30$.
 Thus, I and II both together give the answer. So, correct answer is (e).
3. I. 3 years ago, $\frac{1}{2}(A + B) = 18$
 $\Rightarrow$ 3 years ago, $(A + B) = 36$
 Now, $(A + B) = (36 + 3 + 3) = 42$
 $\Rightarrow A + B = 42$... (i)
 II. Now, $\frac{1}{3}(A + B + C) = 22$
 $\Rightarrow A + B + C = 66$... (ii)
 From (i) and (ii), we get $C = (66 - 42) = 24$.
 Thus, I and II both together give the answer. So, correct answer is (e).
4. I. Reena's present age = $5 \times$ (Her son's present age).
 II. Reena's age 2 years hence = 3 times her daughter's age at that time.
 Clearly, data even in I and II is not sufficient to get Reena's present age.
 $\therefore$ Correct answer is (d).
5. I. $\frac{A}{5} : \frac{B}{4} = 1 : 2 \Leftrightarrow \frac{A}{5} \times \frac{4}{B} = \frac{1}{2}$
 $\Leftrightarrow \frac{A}{B} = \left(\frac{1}{2} \times \frac{5}{4}\right) = \frac{5}{8} \Leftrightarrow A : B = 5 : 8$.
 II. $20B = AB$.
 Let A's age be $5x$ years. Then, B's age is $8x$ years.
 $\therefore 20 \times 8x = 5x \times 8x \Leftrightarrow 40x = 160 \Leftrightarrow x = 4$.
 $\therefore A = 20$ and $B = 32$.
 Thus, I and II together give the answer. So, correct answer is (e).
6. I. Retirement age is 60 years.
 II. There are 50 employees in the department.
 Average age of 50 employees = 30 years.
 Total age of 50 employees = (50×30) years = 1500 years.

Number of employees next year = 40.
 Total age of 40 employees next year
 $= (1500 + 50 - 60 \times 10) = 940$.

$$\text{Average age next year} = \frac{940}{40} \text{ years} = 23\frac{1}{2} \text{ years.}$$

Thus, I and II together give the answer. So, correct answer is (e).

7. I. $F + S = 50$... (i)
 II. $3(F + S) = 5F$... (ii)

$$\text{From II, we get } 2F = 3S \Leftrightarrow \frac{F}{S} = \frac{3}{2}.$$

Thus, II alone gives the answer, but I alone does not give the answer.

$\therefore$ Correct answer is (b).

8. Let Divya's present age be D years and Shruti's present age be S years.
 Then, $D = 2 \times S \Leftrightarrow D - 2S = 0$... (i)

$$\text{I. } \frac{D+5}{S+5} = \frac{9}{5} \text{ ... (ii)}$$

$$\text{II. } \frac{D-10}{S-10} = \frac{3}{1} \text{ ... (iii)}$$

$$\text{From (ii), we get } 5D + 25 = 9S + 45 \Leftrightarrow 5D - 9S = 20 \text{ ... (iv)}$$

$$\text{From (iii), we get } D - 10 = 3S - 30 \Leftrightarrow D - 3S = -20 \text{ ... (v)}$$

Thus from (i) and (iv), we get the answer.

Also, from (i) and (v), we get the answer.

$\therefore$ I alone as well as II alone gives the answer. Hence, the correct answer is (c).

9. I. $A + B = 21$.

$$\text{II. } A - B = 5.$$

$$\text{III. } AB = 104.$$

Clearly, any two of three will give the answer. So, correct answer is (d).

10. I. Let the present ages of Tanya and Rahul be $3x$ years and $4x$ years respectively.

$$\text{II. After 5 years, (Tanya's age) : (Rahul's age) = } 4 : 5.$$

$$\text{III. (Rahul's age) = (Tanya's age) + 5.$$

$$\text{From I and II, we get } \frac{3x+5}{4x+5} = \frac{4}{5}. \text{ This gives } x.$$

$\therefore$ Tanya's age = $3x$ can be found. Thus, I and II give the answer.

From I and III, we get $4x = 3x + 5$. This gives x .

$\therefore$ Tanya's age = $3x$ can be found. Thus, I and III give the answer.

From III : Let Tanya's present age be t years.

Then, Rahul's present age = $(t + 5)$ years.

$$\text{Thus, from II and III, we get : } \frac{t+5}{t+10} = \frac{4}{5}. \text{ This gives } t.$$